

Spatial Proteomics of the Human Small Intestine

Hands-On Exercise

By: Ofra Golani, MICC Cell Observatory, Weizmann Institute of Science **Based on:** CODEX experiment and analysis done by Rouven Hoefflin and published in

Harnik, Yakubovsky, Hoefflin et al. "A spatial expression atlas of the adult human proximal small intestine", Nature 2024

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Section 1: Data Exploration in QuPath

Background

Three patient samples from the duodenum–jejunum junction were imaged using **CODEX** (Co-Detection by indEXing, also called PhenoCycler) with a **37-antibody panel**, resulting in 226,529 segmented and phenotyped cells. The data is stored as a multi-channel image (qptiff format), where each channel corresponds to one antibody or the nuclear DAPI stain.

The Two Tissue Axes

The small intestine is a highly structured organ, composed of repeating crypt–villus units. These structures are polarized by blood flow and gradients of morphogens at their basal side, and by graded microbial and nutrient compositions at their apical sides. Consequently, cells residing in different crypt–villus zones operate in highly variable microenvironments. As a potential adaptation, epithelial, immune and mesenchymal cells have been shown to exhibit substantial spatial variability in their gene and protein expression patterns. The paper characterises this variability along **two perpendicular spatial axes**:

The Crypt–Villus Axis

The **vertical axis** of the villus — from the proliferating crypt at the base to the tip where cells are shed into the lumen. Cells are born in the crypt and migrate upward over several days, progressively changing their protein expression as they go. This axis governs **epithelial zonation** and the distribution of immune and mesenchymal cell types along the villus height.

The Stromal–Epithelial Axis

The **horizontal axis** across the villus width — from the epithelial surface to the stromal core. It determines whether a cell is positioned for direct contact with the epithelium or resides deeper in the lamina propria. One of the questions explored in this paper is whether different immune cells localize differently along this axis.

Keep these two axes in mind as you explore the data.

The CODEX Antibody Panel

The 37 antibodies used in this study are listed below, organized by the cell types they mark. This corresponds to Extended Data Fig. 3a and Supplementary Table 3 in the paper. Note that several markers are included to capture **cell state** (e.g., activation, exhaustion) in addition to cell identity, enabling classification into functionally distinct subtypes — for example, the three enterocyte subtypes defined by GLUT1, NDRG1 and CA9.

Category	Markers	Primary cell types / targets
Pan-immune	CD45	All leukocytes
Macrophage	CD14, CD163, CD206	Monocytes, M2 macrophages
Dendritic cells	CD11c, HLA-DR, CD1c, DC-LAMP	cDC1, cDC2, antigen-presenting cells, activated DCs
T cells	CD3, CD4, CD8, CD69, CD279 (PD-1)	T cells (helper, cytotoxic, activated, exhausted)
B cells / Plasma cells	CD19, CD138	B cells, plasma cells
Fibroblasts / ECM	aSMA, CD90, VCAN, MT1-MMP, FN1	Stromal fibroblasts, basement membrane fibroblasts, ECM
Endothelium	CD31, PDPN	Blood vessels (CD31), lymphatic vessels (PDPN)
Epithelium	GLUT1, NDRG1, CA9, Ki67, POSTN	Enterocyte subtypes (zonated along villus), proliferating cells
Telocytes	PDGFRA, TNFR, TNC, HPLN1	Telocytes, telocyte tip cells, ECM
MALT / Inflammatory	CD24, IL1B	MALT structures, pro-inflammatory cells
Broad / Other	VIM, CD44, GAP43, ACAN	Non-epithelial cells broadly (VIM), stem/stromal activation (CD44), enteric neurons (GAP43), ECM (ACAN)

Note on naming: The marker called **MHCII** in the analysis code corresponds to **HLA-DR** in the image channels and antibody table — these refer to the same target (MHC class II molecule). The file includes few additional channels that were not used (MPO, DCX, CD15, CD83).

Exploring the Raw Data

You will work with **P9_v2_raw** in the project **SmallIntestine_CODEX_QPProj** — the raw CODEX image. Focus on the straight villi structures within pre-selected ROIs (marked as red annotations). Your goal is to develop an intuitive feel for the data before any analysis is applied.

Step 1: Open the project and navigate the image

1. Open **QuPath**. Drag and drop **SmallIntestine_CODEX_QP_Proj\project.qpproj** into the QuPath window to open the project.
2. Double-click **P9_v2_raw** to open it.

3. Use the scroll wheel to zoom in and out. Hold **Space** and drag to pan.
4. At low magnification, locate:
5. The **villi** — finger-like protrusions extending upward
6. The **crypts** — the narrow pockets at the base of each villus
7. The **submucosa** — the layer below the crypts
8. The **red annotations** — pre-selected straight villi ROIs that will be used throughout the exercise

Step 2: Explore individual channels

1. Open the **Brightness & Contrast** panel by clicking the B&C icon in the toolbar (sun-like icon, see QuPath toolbar)
2. You will see a list of all channels, each with an on/off checkbox, a colour swatch, and min/max sliders
3. Turn **all channels off**, then turn them on **one at a time** and observe what you see:

Channel	What to look for
DAPI	All nuclei — gives you the overall tissue architecture
CD45	All immune cells — scattered throughout the lamina propria
CD8	CD8+ T cells — notice how close they are to the epithelial surface
CD4	CD4+ T cells — compare their location to CD8 cells
GLUT1	Epithelial cells — strong signal at the villus surface
PDGFRA	Telocytes — thin layer just beneath the epithelium
CD31	Blood vessels — tubular structures in the stroma
PDPN	Lymphatics — central lacteal in the villus core
Ki67	Proliferating cells — should be restricted to the crypt region
aSMA	Smooth muscle and basement membrane fibroblasts — marks the muscularis mucosae layer

? Observation question: When you look at CD8 vs. CD4, can you already see a difference in their location? What about the difference between CD31 (blood) and PDPN (lymph) vessels — where does each vessel type reside within the villus?

Step 3: Multi-channel overlays — capturing cell type zonation patterns

★ **Tips for channel display:** - Use keyboard numbers 1 – 9 to toggle individual channels on/off quickly - To toggle multiple channels at once: highlight the desired channels in the Brightness & Contrast panel, right-click, and choose show/hide/toggle. Use `Ctrl+A` to select all channels - To change a channel's display colour: double-click on the channel name in the Brightness & Contrast panel - To restore all channels to their original preset colours and settings: in the Brightness & Contrast panel, select **Original_Color_Scheme** from the settings menu

Combination 1 — Immune landscape: Turn on: CD45 , CD8 , CD4 , CD14 - Can you identify CD8 T cells? CD4 T cells? Macrophages?

Combination 2 — Tissue architecture: Turn on: DAPI , GLUT1 , CD31 , PDPN - Identify: epithelium, blood vessels, lymphatic vessel (central lacteal) - Observe the crypt–villus axis: does GLUT1 intensity change along the villus height?

Combination 3 — Epithelial zonation: Turn on: GLUT1 , NDRG1 , CA9 - These three markers define the three enterocyte subtypes in the paper - Do they appear to occupy different positions along the villus?

Combination 4 — Stromal architecture: Turn on: PDGFRA , aSMA , VIM , CD31 - Identify the telocyte layer (thin PDGFRA+ cells just beneath the epithelium), the muscularis mucosae (aSMA+ layer below crypts), and blood vessels (CD31+) - Note: VIM marks all non-epithelial cells broadly, not just fibroblasts

Combination 5 — Dendritic cells: Turn on: DAPI , CD11c , CD1c , HLA-DR - Can you spot dendritic cells? Are they near the epithelium or deeper in the stroma?

Questions

- If you had to choose the panel would you add markers for segmentation, which?
 - You can already see with your eyes that CD8 and CD4 T cells localize differently. Why do we need quantitative analysis if we can see it?
 - Looking at the two tissue axes — how would you measure a cell's position along the crypt–villus axis using the CODEX data? And along the stromal–epithelial axis? Can you think of more than one approach for each?
 - What are the key advantages of spatial proteomics (CODEX) over spatial transcriptomics (Visium) for this kind of study? What does Visium offer that CODEX cannot?
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Section 2: Cell Segmentation

In this section we compare two approaches for segmenting cells in multiplexed fluorescence images: a classical pipeline (nuclear detection with Cellpose followed by cell expansion) and a newer end-to-end method (InstaSeg). By running both on the same tissue region and examining the resulting measurements, we explore how the choice of segmentation strategy can affect downstream quantification of protein expression and cell morphology.

Setup

1. In the Project panel, right-click on **P9_v2** → **Duplicate image** → when prompted, keep the default (**Duplicate with data**) → name it **P9_v2 (1)** → OK.

Opening both images in synchronized mode

1. In QuPath's toolbar, click the **Multi-viewer** button (the icon showing two side-by-side rectangles). This splits the window into left and right viewer panes.
2. Click the **right viewer pane** to make it active (it will gain a highlighted border).
3. In the Project panel, double-click **P9_v2 (1)** to open it in the right pane. The original **P9_v2** remains in the left pane.
4. In the toolbar, click the **Synchronize viewers** button (chain-link icon). Pan and zoom are now locked across both panels — navigating in one mirrors the other.

Left panel — Cellpose: nuclei detection + cell expansion

1. Drag and drop **scripts/cellpose_nuc.groovy** into QuPath to open it in the Script Editor.
2. In the left viewer, hold **Ctrl** and click 1–2 annotation regions over straight villi (they turn yellow when selected).
3. In the Script Editor, click **Run**. This will take a few minutes. The script detects nuclei with Cellpose (DAPI channel) and expands outward to approximate cell boundaries.

Right panel — InstaSeg

1. Click the **right viewer** to make it active.
2. Select the same annotation region(s) as in step 7 (hold **Ctrl** to multi-select).
3. Go to **Extensions** → **InstaSeg**. This opens the InstaSeg dialog.
4. Configure:
 - **Model:** `fluorescence_nuclei_and_cells` (select from the available models, you may be asked to download the model)
 - **Input channels:** select **all channels**

- **Output:** select **all available outputs**; make sure **Make measurements** is ticked
5. Click **Run**. InstaSeg detects nuclei and cell boundaries simultaneously in a single pass, without requiring a separate expansion step.

Comparing the results

1. Because the viewers are synchronized, you can compare both segmentations side by side without switching windows. Display the **DAPI channel** in both viewers to use the nuclei as a reference. Then use the keyboard to toggle the detection overlays:

- Press **D** to toggle all detection outlines on/off
- Press **F** to toggle detection fill on/off

This lets you quickly assess how well the detected cell boundaries correspond to the underlying signal.

2. **Cell measurements** are listed in the **Annotations tab** (bottom-left panel) — make sure you are in that tab to see per-object measurements. Each cell carries intensity measurements per channel (mean, median, min, max, std) for each compartment (nucleus, cytoplasm, cell), as well as **shape measurements** such as area, perimeter, circularity, and solidity.
3. **Measurement maps** let you visually inspect the measurements as colour overlays across all detected cells. Go to **View** → **Show measurement maps**. Use the dropdown to select which measurement to display.

Tip: Measurement maps are most readable in **Fill mode** (press **F** to enable). Adjust the contrast slider to reveal the dynamic range of the measurement you are inspecting.

Suggested measurements to explore: - **Shape Measurements: Cell: Area / Cell: Circularity / Cell Length** — compare the distributions between the two methods - **Cell: CD4 / CD8 mean** — immune cell markers. To inspect these: enable the **CD4** and **CD8** fluorescence channels in the viewer, then toggle the measurement map overlay on and off to compare the raw signal with the quantified per-cell values. - **Cell: GLUT1 / NDRG1 / CA9 mean** — does it capture the difference in expression along the villi?

Questions

- Which method detects more cells in the selected region? How different is the segmentation?
- How do differences in cell boundary delineation between the two methods affect the intensity measurements? Consider a marker that is expressed at the cell membrane — would the two methods give the same mean value?
- Do you see cases of signal leakage from neighbouring cells?

- What differences do you observe between different cell types (e.g. columnar enterocytes versus small round immune cells) in terms of segmentation quality and measurement values?
 - Based on your inspection, which method would you prefer for this tissue type, and why?
 - Which method was used in the paper?
 - Here we used InstaSeg with **all channels** as input. Does this make sense? Would you select otherwise?
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Section 3: Exploring Cell Types and their Spatial Organization

In this section we will use the analyzed data deposited with the paper and explore how the 18 classified cell types are distributed across the intestinal tissue, focusing on the spatial zonation of epithelial and immune populations.

1. Open **P9_v2_analyzed**. Press **F** to enable Fill mode and view the 18 cell types in their assigned colors.
 2. In the **Classes panel**, hide all classes except `Epi_CA9`, `Epi_NDRG1`, `Epi_GLUT1`, `Tcell_CD4`, `Tcell_CD8`, and `fib_BM`. (Use the eye icon on the right side of each class.)
 3. Navigate to a region with well-oriented straight villi and inspect the spatial distribution of these populations. Compare what you see with **Figure 4b and 4c** from the paper.
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Questions

- **Figure 4b** shows the zonation of cell types along the **crypt–villus axis**, and **Figure 4c** shows zonation along the **stromal–epithelial axis**. How was each of these axes calculated from the spatial data?
- How were the cell type labels generated in the paper?
- Given that you start from a segmented image with per-cell intensity and shape measurements, what steps are needed to arrive at a classification of 18 distinct cell types?